Discontinuous Function

At a point of discontinuous at $x = c$.

$$f(x) = \frac{1}{2}[f(c-0) + f(c+0)]$$

where, $f(c-0) = LHS$ of $f(x)$ at $x = c$
& $f(c+0) = RHS$ of $f(x)$ at $x = c$.

Let $f(x)$ be defined as,

$$f(x) = \begin{cases} f_1(x), & x_0 < x < c \\ f_2(x), & c < x < (x_0 + 2\pi) \end{cases}$$

where, c is the ~~piece~~ point of discontinuity in $(x_0, x_0 + 2\pi)$

Then, $$a_0 = \frac{1}{\pi}\left[\int_{x_0}^{c} f_1(x)\,dx + \int_{c}^{(x_0+2\pi)} f_2(x)\,dx\right]$$

$$a_n = \frac{1}{\pi}\left[\int_{x_0}^{c} f_1(x)\cos nx\,dx + \int_{c}^{(x_0+2\pi)} f_2(x)\cos nx\,dx\right]$$

$$b_n = \frac{1}{\pi}\left[\int_{x_0}^{c} f_1(x)\sin nx\,dx + \int_{c}^{(x_0+2\pi)} f_2(x)\sin nx\,dx\right]$$

$$\Rightarrow f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx.$$

Q.1) Find the fourier series for $f(x)$ if,

$$f(x) = \begin{cases} -\pi & -\pi < x < 0 \\ x & 0 < x < \pi \end{cases}$$

deduce that: $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$

Sol^n:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Now,

$$a_0 = \frac{1}{\pi}\left[\int_{-\pi}^{0} f(x)\,dx + \int_{0}^{\pi} f(x)\,dx\right]$$

$$= \frac{1}{\pi}\left[\int_{-\pi}^{0} (-\pi)\,dx + \int_{0}^{\pi} x\,dx\right]$$

$$= \frac{1}{\pi}\left[-\pi\left[x\right]_{-\pi}^{0} + \left[\frac{x^2}{2}\right]_{0}^{\pi}\right]$$

$$= -(0-(-\pi)) + \frac{1}{\pi}\left(\frac{\pi^2}{2} - 0\right)$$

$$= -\pi + \frac{1}{\pi} \times \frac{\pi^2}{2}$$

$$\Rightarrow a_0 = -\pi + \frac{\pi}{2} = \frac{-\pi}{2}.$$

$$a_n = \frac{1}{\pi}\left\{\int_{-\pi}^{0} f(x)\cos nx\,dx + \int_{0}^{\pi} f(x)\cos nx\,dx\right\}$$

$$= \frac{1}{\pi}\int_{-\pi}^{0} -\pi.\cos nx\,dx + \frac{1}{\pi}\int_{0}^{\pi} x.\cos nx\,dx$$

$$= -\int_{-\pi}^{0}\cos nx\,dx + \frac{1}{\pi}\left\{\left[\frac{x.\sin nx}{n}\right]_0^{\pi} - \int_0^{\pi}\frac{(1)\sin nx}{n}\,dx\right\}$$

$$= -\left[\frac{\sin nx}{n}\right]_{-\pi}^{0} + \frac{1}{\pi}\times 0 + \frac{1}{\pi}\left[\frac{\cos nx}{n^2}\right]_0^{\pi}$$

$$\Rightarrow a_n = \frac{1}{\pi n^2}\left[(-1)^n - 1\right]$$

Also,

$$b_n = \frac{1}{\pi}\left\{\int_{-\pi}^{0}(-\pi).\sin nx\,dx + \int_0^{\pi} x.\sin nx\,dx\right\}$$

$$= -\int_{-\pi}^{0}\sin nx\,dx + \frac{1}{\pi}\left\{\left[x.\frac{-\cos nx}{n}\right]_0^{\pi} - \int_0^{\pi}\frac{-\cos nx}{n}\,dx\right\}$$

$$= \left[\frac{\cos nx}{n}\right]_{-\pi}^{0} - \frac{1}{\pi n}\left[\pi.(-1)^n - 0\right] + \frac{1}{\pi}\left[\frac{\sin nx}{n^2}\right]_0^{\pi}$$

$$= \frac{1-(-1)^n}{n} - \frac{(-1)^n}{n} + \frac{1}{\pi}\times 0$$

$$\Rightarrow b_n = \frac{1-2(-1)^n}{n}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

$$\Rightarrow f(x) = \frac{-\pi}{4} + \frac{1}{\pi} \sum_{n=1}^{\infty} \left(\frac{(-1)^n - 1}{n^2} \right) \cos nx$$

$$+ \sum_{n=1}^{\infty} \left(\frac{1 - 2(-1)^n}{n} \right) \sin nx$$

$$f(0) = \frac{-\pi}{4} + \frac{1}{\pi} \sum_{n=1}^{\infty} \left(\frac{(-1)^n - 1}{n^2} \right) \cos 0 + \sum_{n=1}^{\infty} \left(\frac{1 - 2(-1)^n}{n} \right) \sin 0$$

$$= \frac{-\pi}{4} + \frac{1}{\pi} \left(\frac{-2}{1} + 0 - \frac{2}{3^2} + 0 - \frac{2}{5^2} + \ldots \right) + 0$$

$$= \frac{-\pi}{4} - \frac{2}{\pi} \left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots \right)$$

$$f(0) = \frac{1}{2} [f(0-0) + f(0+0)]$$

$$f(0-0) = \lim_{h \to 0} f(0-h) = -\pi$$

$$f(0+0) = \lim_{h \to 0} f(0+h) = 0$$

$$f(0) = \frac{1}{2}(-\pi + 0) = \frac{-\pi}{2}$$

$$\Rightarrow \frac{-\pi}{2} = \frac{-\pi}{4} - \frac{2}{\pi}\left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots\right]$$

$$\Rightarrow \frac{-\pi}{2} + \frac{\pi}{4} = \frac{-2}{\pi}\left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots\right]$$

$$\Rightarrow \frac{-\pi}{4} = \frac{-2}{\pi}\left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots\right]$$

$$\Rightarrow \frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$$ proved

2. Find the fourier series expression for $f(x)$ if,

$$f(x) = \begin{cases} 0 , & -\pi \leq x \leq 0 \\ \sin x , & 0 \leq x \leq \pi \end{cases}$$

Prove that: $f(x) = \frac{1}{\pi} + \frac{\sin x}{2} - \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{\cos 2n\pi}{4n^2 - 1}$

Hence show that: $\frac{1}{1.3} - \frac{1}{3.5} + \frac{1}{5.7} - \dots - \infty = \frac{1}{4}(\pi - \infty)$

Sol^n:

The solution is,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx \quad \text{—(1)}$$

Now,

$$a_0 = \frac{1}{\pi} \left\{ \int_{-\pi}^{0} f(x)\,dx + \int_{0}^{\pi} f(x)\,dx \right\}$$

$$= \frac{1}{\pi} \left[\int_{-\pi}^{0} 0\,dx + \int_{0}^{\pi} \sin x\,dx \right]$$

$$= \frac{1}{\pi} \left[\left[-\cos x \right]_{0}^{\pi} \right]$$

$$= \frac{1}{\pi} \left[-\cos \pi + \cos 0 \right]$$

$$= \frac{1}{\pi} (1+1)$$

$$\Rightarrow a_0 = \frac{2}{\pi}$$

Then,

$$a_n = \frac{1}{\pi}\left[\int_{-\pi}^{0} f(x) \cos nx \, dx + \int_0^{\pi} f(x) \cos nx \, dx\right]$$

$$= \frac{1}{\pi} \int_0^{\pi} \sin x . \cos nx \, dx$$

$$= \frac{1}{2\pi} \int_0^{\pi} 2 \sin x . \cos nx \, dx$$

$$= \frac{1}{2\pi} \int_0^{\pi} [\sin(1+n)x + \sin(1-n)x] \, dx$$

$$= \frac{1}{2\pi} \left[\frac{-\cos(1+n)x}{(1+n)} - \frac{\cos(1-n)x}{(1-n)}\right]_0^{\pi}$$

$$= \frac{1}{2\pi}\left[\frac{-\cos(1+n)\pi}{(1+n)} - \frac{\cos(1-n)\pi}{(1-n)} + \frac{\cos 0}{(1+n)} + \frac{\cos 0}{(1-n)}\right]$$

$$= \frac{1}{2\pi}\left[\frac{-(-1)^{1+n}}{(1+n)} - \frac{(-1)^{1-n}}{(1-n)} + \frac{1}{(1+n)} + \frac{1}{(1-n)}\right]$$

$$= \frac{1}{2\pi}\left[\frac{1-(-1)^{1+n}}{1+n} + \frac{1-(-1)^{1-n}}{1-n}\right]$$

$$\boxed{2\sin A \cdot \sin B = \cos(A-B) - \cos(A+B)}$$

When n is odd,

$$a_n = 0$$

When n is even,

$$a_n = \frac{-2}{\pi(n^2-1)}.$$

Also,

$$b_n = \frac{1}{\pi}\int_0^{\pi} \sin x \cdot \sin nx \, dx$$

$$= \frac{1}{2\pi}\int_0^{\pi} [\cos(1-n)x - \cos(1+n)x] \, dx$$

$$= \frac{1}{2\pi}\left[\frac{\sin(1-n)x}{(1-n)} - \frac{\sin(1+n)}{(1+n)}\right]_0^{\pi}$$

$$\Rightarrow b_n = 0 \quad [n \neq 1]$$

$$b_1 = \frac{1}{\pi}\int_0^{\pi} \sin x \cdot \sin x \, dx$$

$$= \frac{1}{\pi}\int_0^{\pi}\left(\frac{1-\cos 2x}{2}\right) dx$$

$$= \frac{1}{2\pi}\left[x - \frac{\sin 2x}{2}\right]_0^{\pi}$$

$$= \frac{1}{2\pi}[\pi - 0]$$

$$\Rightarrow b_1 = \frac{1}{2}$$

Hence, eqⁿ. (1) becomes,

$$f(x) = \frac{1}{\pi} - \frac{2}{\pi}\left(\frac{\cos 2x}{2^2-1} + \frac{\cos 4x}{4^2-1} + \ldots\right)$$

$$+ \frac{1}{2}\sin x$$

$$\Rightarrow f(x) = \frac{1}{\pi} + \frac{\sin x}{2} - \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{\cos 2n\pi}{4n^2-1}$$ proved